

## Unit 2, Lesson 5 – Implicit Differentiation

When you look at most of the functions we've taken derivatives of so far, they are clearly defined with  $y$  as a function of  $x$ . To put it another way, they have the form  $y = f(x)$ . Because of this clarity, we call these types of functions *explicitly defined functions*, or just **explicit functions**.

However, not all relations between variables are so clear-cut.

For example, consider the relation  $x^2 + y^2 = 16$ . What does the graph of this relation look like?

This relation does **not** define a single function  $y = f(x)$  over its entire graph. However, we *can* solve for  $y$  locally as one of two functions. For this type of relation, we are **implicitly** treating  $y$  as a function of  $x$ . Thus, equations in which  $x$  and  $y$  are related without one variable being explicitly isolated are called **implicit equations** or **implicit relations**, and the method that we use to differentiate them is known as **implicit differentiation**. It's just an extension of the Chain Rule.

What would we get if we solved  $x^2 + y^2 = 16$  for  $y$ ? How do the graphs of these functions relate to the graph of the original relation?

One of the useful features of implicit equations is that we do not need to solve explicitly for  $y$  before differentiating. Near a particular point, the relation may define  $y$  as a function of  $x$ , even when solving explicitly for  $y$  is difficult, inconvenient, or impossible. We can then differentiate the equation while keeping  $y$  in the expression.

### GUIDELINES FOR IMPLICIT DIFFERENTIATION

1. Differentiate both sides of the equation *with respect to*  $x$ .
2. Collect all terms involving  $dy/dx$  on the left side of the equation and move all other terms to the right side of the equation.
3. Factor  $dy/dx$  out of the left side of the equation.
4. Solve for  $dy/dx$ .

When differentiating with respect to  $x$ , expressions involving only  $x$  behave as usual. When we differentiate an expression involving  $y$ , we must remember that  $y$  itself is a function of  $x$ , so that the Chain Rule introduces a factor of  $\frac{dy}{dx}$ .

**Example 1.** Calculate both derivatives.

a.  $\frac{d}{dx}[3x^2]$

b.  $\frac{d}{dx}[3y^2]$

c.  $\frac{d}{dx}[x + 3y]$

d.  $\frac{d}{dx}[xy^2]$

Here, notice that we are not actually finding  $\frac{dy}{dx}$ , we are merely expressing the derivative of the particular expression in terms of  $\frac{dy}{dx}$ .

We can, however, actually *solve* for  $\frac{dy}{dx}$  when we have an equation involving  $x$  and  $y$ .

**Example 2.** Find  $\frac{dy}{dx}$  given that  $y^3 + y^2 - 5y - x^2 = -4$ .

Now that we have the derivative, let's calculate the slope of the function at the following points:

a.  $(2,0)$

b.  $(1,-3)$

c.  $x = 0$

d.  $(1,1)$

We can also examine this using Desmos. Why do some calculated slopes require both an  $x$ -coordinate and a  $y$ -coordinate to calculate properly?

Before we do more algebra, let's ask when an implicit relation actually behaves like a function.

**Example 3.** If possible, represent  $y$  as a differentiable function of  $x$ .

a.  $x^2 + y^2 = 0$

b.  $x^2 + y^2 = 1$

c.  $x + y^2 = 1$

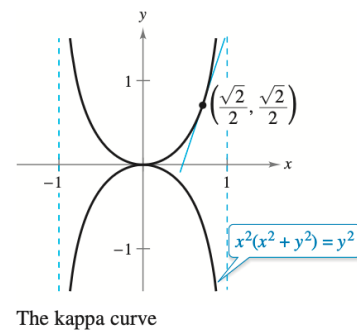
**Example 4.** Determine the slope of the tangent line to the graph of  $x^2 + 4y^2 = 4$  at the point  $(\sqrt{2}, -\frac{1}{\sqrt{2}})$ .

**Example 5.** Determine the slope of the graph of  $3(x^2 + y^2)^2 = 100xy$  at the point  $(3,1)$ .

**Example 6\*.** Find  $\frac{dy}{dx}$  implicitly for the equation  $\sin y = x$ . Then find the largest interval of the form  $-a < y < a$  on which  $y$  is a differentiable function of  $x$ .

**Example 7.** Given  $x^2 + y^2 = 25$ , find  $\frac{d^2y}{dx^2}$ .

**Example 8.** Find the tangent line to the graph of  $x^2(x^2 + y^2) = y^2$  at the point  $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$  as shown in the figure below.



## Practice

A. Mechanics. *Differentiate, but don't solve for  $y'$  yet.*

1.  $\frac{d}{dx}[y^4]$

2.  $\frac{d}{dx}[x^2y^3]$

3.  $\frac{d}{dx}[\sin y]$

4.  $\frac{d}{dx}[\cos(x + y)]$

B. Basic implicit differentiation. *Find  $\frac{dy}{dx}$ .*

5.  $x^2 + y^2 = 36$

6.  $x^2 + y^3 = 6xy$

7.  $x^2 + xy + y^2 = 7$

8.  $x^2y + y^2 = 10$

C. Find a slope or tangent line.

9. Find the slope of the tangent line to  $x^2 + 4y^2 = 20$  at  $(2,2)$ .

10. Find the equation of the tangent line to  $x^3 + y^3 = 9xy$  at  $(3,3)$ .

11. The curve  $x^2 + xy + y^2 = 7$  contains the point  $(1,2)$ . Find the tangent line at the point.

D. Questions that require interpretation.

12. For  $x^2 + y^2 = 25$ , find all points where the tangent line is horizontal.

(Follow up) For the same curve, find all points where the tangent line is vertical.

13. For  $x + y^2 = 4$ , explain why  $y$  cannot be represented by a single differentiable function of  $x$  on the entire relation, but *can* be represented locally by two differentiable functions.

E. More sophisticated chain/product rule combinations. Find  $\frac{dy}{dx}$ .

14.  $\sin(x + y) = xy$

15.  $x^2 \sin y + y^2 = 4$

16.  $x \sin y + y \cos x = 3$

F. Higher-order/extension.

17. Given  $x^2 + y^2 = 16$ , find  $\frac{d^2y}{dx^2}$ .

18. If  $xy = 6$ , find  $\frac{d^2y}{dx^2}$ .

19. If  $\cos y = x$ , find  $y'$  and determine where  $y$  can be a differentiable function of  $x$ .